

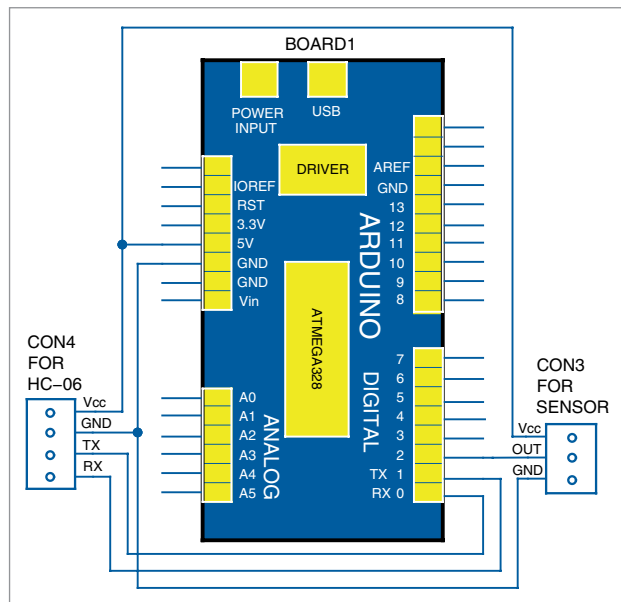


Fig. 3: Circuit diagram for speed monitoring on smartphone

software.

Select the correct board from Tools → Board menu in Arduino IDE and burn the program (sketch) through the standard USB port in your computer. Here, *Serial.begin(9600)* function

initialises the serial port with a baud rate of 9600. The *attachInterrupt(0, counter, FALLING)* function calls the interrupt handler “counter” whenever a signal connected to interrupt pin 0 of the Arduino makes a high-to-low, that is, falling edge transition.

After uploading the code to Arduino board (Board1), connect HC-06 Bluetooth to CON4 and sensor digital output to CON3. Mount the disc encoder wheel on the shaft of the DC motor. Place the disc encoder between the groove of the opto-sensor, as shown in the prototype above.

Pair the Bluetooth between the smartphone and HC-06. Open the

Arduino Bluetooth Controller app from an Android phone and select Terminal Mode. Now switch on the motor. As the encoder disc rotates between the optical encoder groove, the RPM of the motor can be read on the smartphone. The RPM can also be read on the serial monitor of Arduino. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com

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Souvik Kumar Das is passionate about electronics and MCU-based system design



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